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# A Unified Approach to Motion and Force Control of Flexible Joint Robots †

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## Abstract

*This paper addresses motion and force control issues of flexible joint robots in constrained motion in the presence of parametric uncertainty. Assuming that values of all system parameters, including joint flexibility, are unknown, except for some of their bounds, a new adaptive and sliding mode control scheme is developed, that provides a unified approach to motion and force control of flexible joint robots in general coordinate space. The proposed control scheme, consisting of an adaptive constrained motion/force controller and a joint torque sliding mode controller, guarantees that motion, force and joint torque tracking errors converge asymptotically to zero without persistent excitation. Simulation results illustrate the effectiveness of the proposed control method.*

## 1. INTRODUCTION

Many advanced robot applications such as assembly and manufacturing require mechanical interaction of the robot manipulator with the environment. For successful execution of such tasks, it is desired that both motion and contact force be controlled simultaneously [1, 2, 3]. In many applications, the contacted environment is rigid, and the displacement of the end-effector in the direction normal to the contacted surface is constrainted to be zero. In this case, the robot motion is defined as constrained motion. Much attention has been given to control of robots in constrained motion. A general theoretical framework of constrained motion control was developed by McClamroch and Wang [4]. A reduced dynamic model was formulated to represent constrained robot dynamics, and a modified computed torque control method was proposed. In the work of Mills and Goldenberg [5], descriptor theory has been applied to constrained motion control and a linearized feedback controller was developed. A number of adaptive control algorithms and robust control methods have been also proposed to deal with the case of parametric uncertainty [6, 7, 8].

In the work cited above, the robot manipulator is modeled as a completely rigid structure. So far, only a few studies have addressed the force control of flexible joint robots. The work [9, 10] has pointed out that control of robots based on rigid-body dynamics formulation is inadequate in dealing with more stringent operating conditions. Joint flexibility should be taken into account in modeling and control design if high performance is to be achieved. Spong [11] addressed the force control issue of flexible joint robots and derived a control algorithm for both hybrid control and impedance control methods. Mills [12] developed a control method by using a composite control technique on a singular perturbation model of flexible joint robots in constrained motion. Jankowski and ElMaraghy [13] proposed a nonlinear decoupling control method for constrained robots with flexible joints. All these results are based on the exact knowledge of the dynamic model. Recently, an adaptive force control scheme for a single-link mechanism with joint flexibility was developed by Lian, Jean and Fu [14]. In their work, explicit force control approach is used, but constrained motion control is not considered, and the method can not be easily extended to the general n-link case. Based on the reduced dynamic model, an adaptive and sliding mode two-stage control scheme was proposed by Lin and Goldenberg [16] for control of flexible joint robots in constrained motion. The control scheme was established in a systematic way for flexible joint robots in the general n-link case, without requiring the exact knowledge of robot manipulator parameters.

Usually, control design based on reduced dynamic model, uses only the motion tracking error in its adaptive mechanism. Therefore, only motion tracking is achieved and force tracking error can only be guaranteed to be bounded. Force error converges to zero if persistent excitation condition is satisfied [6, 16]. For rigid robots in constrained motion, some methods have been proposed to achieve motion and force tracking without the requirements of persistent excitation [15, 8]. In this paper, a new global convergent control scheme for flexible joint robots in constrained motion is developed by generalizing our previous control strategy [16]. It has two important features: i) it is proved theoretically that, with the proposed new control scheme, motion, force and joint torque tracking errors converge simultaneously to zeros without persistent excitation; and ii) the control scheme provides a unified approach for motion and force control of flexible joint robots in general coordinate frame (including task space or joint space). Furthermore, friction force on the contact surface can be compensated by the control scheme, thus the frictionless contact assumption is no longer needed. Also, distinct from the work of [15], the restriction that the force feedback gain must be very small is not required in the proposed control algorithm. In the following, the constrained dynamic model of flexible joint robots is presented in Section 2. The control scheme and convergence analysis are developed in Section 3. Simulation examples are given in Section 4 to illustrate the effectiveness of the proposed

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control methods. Conclusions are given in Section 5.

## 2. CONSTRAINED DYNAMIC MODEL OF FLEXIBLE JOINT ROBOTS

Consider the dynamic equations of a non-redundant n-link flexible joint robot with joint torque measurements described as follows [10, 17]:

$$D(q_l)\ddot{q}_l + C(q_l,\dot{q}_l)\dot{q}_l + G(q_l) = \tau_s + f \quad (1)$$

$$I_m\ddot{q}_m + B_m\dot{q}_m + \tau_s = u \quad (2)$$

and

$$\tau_s = K_s(q_m - q_l) \quad (3)$$

where $D(q_l)$ is the $n \times n$ positive definite, symmetric inertia matrix; $C(q_l, \dot{q}_l)$ is the $n \times 1$ vector containing coriolis, centrifugal terms; $G(q_l)$ is the $n \times 1$ vector of gravitational forces; $I_m$ is the $n \times n$ constant diagonal matrix with diagonal elements $\gamma_i(\gamma_i + 1)J_{mi}$, $J_{mi}$ is the inertia of the rotor/gear; $B_m$ is the $n \times n$ diagonal matrix of damping terms and $K_s$ is a $n \times n$ diagonal matrix of the joint torsional stiffness. $q_l$ and $q_m$ are n dimensional vectors, represent the link angles and rotor angles respectively. $u$ is the $n \times 1$ input torque control vector, $\tau_s$ is the vector of joint torque measurements and $f$ is the vector of contact force in joint space.

Denoting $F$ to be the contact force in task space, we have

$$f = J^T F$$

where $J \in R^{n\times n}$ is the Jacobian matrix which is assumed to be nonsingular in the defined work space.

Let us consider that the environmental constraints are described by a set of algebraic equations

$$\Phi(X) = 0, \quad \Phi(X) = [\phi_1(X), \cdots, \phi_m(X)]^T \; m \leq n \quad (4)$$

where the mapping $\Phi$: $R^n \rightarrow R^m$ is twice continuously differentiable. The contact force $F$ in task space can be written as [15]

$$F = F_n + F_t = (\eta^T(X) + \xi^T(\mu, X, \dot{X}))\lambda \quad (5)$$

where $\lambda \in R^m$ is the Lagrangian multiplier,

$$F_n = \eta^T(X)\lambda \in R^n, \quad \eta(X) = \frac{\partial \Phi}{\partial X} \in R^{m\times n}$$

represents the constraint force, i.e. normal contact force in the cartesian space, and

$$F_t = \xi^T(\mu, X, \dot{X})\lambda \in R^n, \quad \xi \in R^{m\times n}$$

is the friction force in the tangential direction of the surface. The magnitudes depend on $F_n$, i.e., $\lambda$, and friction coefficient $\mu$, with the sign determined by the end-effector velocity. $\xi$ is linear w.r.t. $\mu$. In the assumption of frictionless contact surface, $F_t = 0$. For constrained motion of robots, the constraint force $F_n$ need to be controlled.

Since the presence of $m$ constraints causes the robot to lose $m$ degrees of freedom, only $(n - m)$ position coordinates can be specified independently, i.e., only $(n - m)$ position coordinates of the robot need to be controlled in the constrained motion. Properly choosing $(n - m)$ linearly independent curvilinear coordinates $\Psi(X) = [\psi_1(X), \cdots, \psi_{n-m}(X)]^T$, combining with the constraints (4), the motion of the robot is then uniquely determined. Let's define the general curvilinear coordinates as

$$r = \begin{bmatrix} r_p \\ r_f \end{bmatrix} \quad (6)$$

with

$$r_p = \Psi(X) = [\psi 1(X), \cdots, \psi_{n-m}(X)]^T \in R^{n-m}$$

$$r_f = \Phi(X) = [\phi_1(X), \cdots, \phi_m(X)]^T = 0 \in R^m$$

Differentiate (6), yields

$$\dot{r} = J_X\dot{X} = J_X J\dot{q}_l = J_q\dot{q}_l \quad (7)$$

where

$$J_X = \frac{\partial r(X)}{\partial X} = \begin{bmatrix} \frac{\partial \Psi}{\partial X} \\ \frac{\partial \Phi}{\partial X} \end{bmatrix} = \begin{bmatrix} \frac{\partial \Psi}{\partial X} \\ \eta(X) \end{bmatrix} \in R^{n\times n}$$

$$J_q = \frac{\partial r(X(q_l))}{\partial q_l} = \frac{\partial r(X)}{\partial X}\frac{\partial X(q_l)}{\partial q_l} = J_X J \in R^{n\times n}$$

Denoting a transformation matrix $T(r) = J_q^{-1}$, we have

$$\dot{q}_l = J_q^{-1}\dot{r} = T(r)\dot{r}$$

$$\ddot{q}_l = T(r)\ddot{r} + \dot{T}(r)\dot{r}$$

Substituting above relations into (1), we get

$$D(r)T\ddot{r} + [D(r)\dot{T} + C(r,\dot{r})T]\dot{r} + G(r) = \tau_s + f \quad (8)$$

Multiplying from left by $T^T$, equation (8) can be rewritten in the compact form as

$$A(r)\ddot{r} + B(r,\dot{r})\dot{r} + W(r) = T^T\tau_s + T^T f \quad (9)$$

where

$$A(r) = T^T D(r)T$$

$$B(r,\dot{r}) = T^T[D(r)\dot{T} + C(r,\dot{r})T] = T^T H(r,\dot{r})$$

$$W(r) = T^T G(r)$$

Thus, using the transformation matrix $T$, the constrained dynamic equation is derived from the dynamic equation (1) with the constraints (4). This equation possesses the following properties:

*Property 1.* By similarity transformation, matrix $A$ is still a bounded positive definite, symmetric matrix, and

$$\dot{A}(r) - 2B(r,\dot{r}) = T^T(\dot{D} - 2C)T \quad (10)$$

is skew symmetric.

*Property 2.* The LHS of equation (9) can be expressed as

$$A(r)\ddot{r} + B(r,\dot{r})\dot{r} + W(r) = Y_r(r,\dot{r},\ddot{r})P \quad (11)$$

where $Y_r(.)$ is an $n \times i$ matrix of known functions, referred to as the regressor, and $P$ is an $i$-dimensional vector of parameters.

*Property 3.* Since $\Phi(X) = \Phi(X(q_l)) \equiv 0$, it is evident that $\eta(X)J(q_l)\dot{q}_l = 0$. If we partition $T = [L, M]$, then $\dot{q}_l = T\dot{r} = L\dot{r}_p$, and we have

$$\eta(X)J(q_l)L\dot{r}_p = 0$$



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As $r_p$ are linearly independent coordinates, it yields that

$$\eta(X)J(q_l)L = 0, \quad L^T J^T \eta^T(x) = 0$$

and

$$L^T J^T \eta^T(x)\lambda = L^T J^T F_n = 0$$

or

$$T^T J^T F_n = \begin{bmatrix} L^T \\ M^T \end{bmatrix} J^T F_n = \begin{bmatrix} 0 \\ M^T J^T \eta^T(x)\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ f_n \end{bmatrix}$$

This means that by similarity transformation, corresponding to $(n-m)$ independent position coordinates $r_p$, the constraint force $F_n$ can always be transformed into $m$ independent components to be controlled. The $(n-m)$ independant coordinates $r_p$, combined with $m$ independant force components $f_n = M^T J^T \eta^T(x)\lambda$ span the n-dimensional vector space.

The above properties are essential for the motion/force controller developed in the next section.

With joint torque measurements $\tau_s$, from equation (3), we have

$$q_m = K_s^{-1}\tau_s + q_l$$

Substituting this equality into equation (2), we get

$$I_{ts}\ddot{\tau}_s + B_{ts}\dot{\tau}_s + h_{ts}(r, \dot{r}, \ddot{r}, \tau_s) = u \tag{12}$$

where

$$I_{ts} = I_m K_s^{-1}$$
$$B_{ts} = B_m K_s^{-1}$$
$$h_{ts} = I_m \ddot{q}_l + B_m \dot{q}_l + \tau_s = I_m T\ddot{r} + (I_m \dot{T} + B_m T)\dot{r} + \tau_s$$

Equations (9) and (12) constitute a dynamic representation of the flexible joint robot in constrained motion.

It is obvious that the derived constrained dynamic model is a general model because the choice of $r_p = \Psi(X)$ is flexible. By properly chosing $r_p$, the model can be expressed in generalized joint or task spaces. Hence, the dynamic model gives a unified expression for flexible joint robots in constrained motion, and consequently, the proposed control scheme developed in Section 3 provides a unified approach for motion and force control of flexible joint robots.

## 3. ADAPTIVE AND SLIDING CONTROL SCHEME

The objective of constrained robot control is to determine the input torque to achieve trajectory tracking on the constrained surface with specified constraint force. Based on the two-stage control strategy [16], a control scheme consisting of a constrained motion/force controller and a joint torque controller is developed in such a way that the constrained motion/force controller is designed to generate the *"desired joint torque"* $\tau_{sd}$, and the joint torque controller computes the required control torque $u$ so that joint torque $\tau_s$ can follow the desired joint torque $\tau_{sd}$, and thus the whole system achieves the control objective.

### 3.1 Adaptive Constrained Motion/Force Controller

Let us define tracking errors:

$$e_p = r_p - r_{pd}$$
$$e_f = \lambda - \lambda_d$$
$$e_t = \tau_s - \tau_{sd}$$

where $r_{pd} = \Psi(X(q_d(t))) \in R^{n-m}$ is the desired robot motion trajectory, and $r_{pd} \in C^4$ is bounded; $\lambda_d \in R^m$ is the desired constraint force trajectory. Define a n-dimensional vector $S$ as

$$S = \begin{bmatrix} \dot{e}_p + k_p e_p \\ k_f \int_0^t e_f dt \end{bmatrix} = \begin{bmatrix} S_p \\ S_f \end{bmatrix} \tag{13}$$

and denote

$$v_r = \dot{r} - S = \begin{bmatrix} \dot{r}_{pd} - k_p e_p \\ -k_f \int_0^t e_f dt \end{bmatrix} \tag{14}$$

The following motion and force controller is proposed to generate the desired joint torque $\tau_{sd}$

$$T^T \tau_{sd} = \hat{A}(r)\dot{v}_r + \hat{B}(r, \dot{r})v_r + \hat{W} - T^T f - (T^T K_D T)S - K_{fD}\dot{S}_f \tag{15}$$

or

$$\tau_{sd} = T^{-T}(T^T \tau_{sd}) \tag{16}$$
$$= \hat{D}T\dot{v}_r + \hat{H}v_r + \hat{G} - f - K_D TS - T^{-T}K_{fD}\dot{S}_f$$

where $K_{fD} = [0, k_{fD}]^T$. $k_p$, $k_f$, $K_D$ and $k_{fD}$ are diagonal matrices of positive gains. $\hat{A}$, $\hat{B}$ and $\hat{W}$ are matrices with the estimated parameters.

Substituting (15) into (9) leads to the following closed loop dynamic equations:

$$A\dot{S} + BS = T^T e_t + (\hat{A} - A)\dot{v}_r + (\hat{B} - B)v_r + (\hat{W} - W) - (T^T K_D T)S - K_{fD}\dot{S}_f$$
$$= T^T e_t + Y_r(r, \dot{r}, v_r, \dot{v}_r)\tilde{P} - (T^T K_D T)S - K_{fD}\dot{S}_f \tag{17}$$

with $\tilde{P} = \hat{P} - P$. The parameter update law is given as

$$\dot{\hat{P}} = -\Gamma^{-1} {Y_r}^T(r, \dot{r}, v_r, \dot{v}_r)S \tag{18}$$

It is noted that the parameter estimation is updated by both motion and force tracking errors $S$.

### 3.2 Joint Torque Sliding Mode Controller

Joint torque controller is designed to drive the actual joint torque to follow the desired joint torque $\tau_{sd}$. To deal with the uncertainty in the rotor subsystem, sliding mode control approach is used. We define the sliding surface as

$$s_t = \dot{\tau}_s - \dot{\tau}_{sd} + \Lambda_t(\tau_s - \tau_{sd}) = \dot{e}_t + \Lambda_t e_t \tag{19}$$

and denote

$$v_t = \dot{\tau}_s - s_t = \dot{\tau}_{sd} - \Lambda_t e_t$$

For tracking the desired joint torque $\tau_{sd}$, the joint torque sliding control law is given as [16]:

$$u = \hat{I}_{ts}\dot{v}_t + \hat{B}_{ts}v_t + \hat{h}_{ts} - K_{tD}s_t - k_t sgn(s_t) \tag{20}$$

where $\hat{I}_{ts}$, $\hat{B}_{ts}$ and $\hat{h}_{ts}$ are the estimated nominal parameters. $K_{tD}$ and $\Lambda_t$ are diagonal positive definite gain matrices, $k_t$ is a positive definite gain matrix with minimum eigenvalue $\lambda_{k_t}^{min}$.



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Substituting this control law into (12), yields:

$$I_{ts}\dot{s}_t + B_{ts}s_t + K_{tD}s_t = w(\dot{r}, \ddot{r}, \tau_s, v_t, \dot{v}_t) - k_t sgn(s_t) \quad (21)$$

where $w = (\hat{I}_{ts} - I_{ts})\dot{v}_t + (\hat{B}_{ts} - B_{ts})v_t + (\hat{h}_{ts} - h_{ts})$, represents the dynamic uncertainty in the rotor subsystem. It is bounded and can be expressed as

$$\| w \| \leq \varrho \quad (22)$$

### 3.3 Convergence Analysis

The stability and convergence properties of the entire control system are analyzed based on Lyapunov stability theory. With the proposed control scheme, we have the following theorem:

**Theorem 1** *Consider the robot dynamic system (9) and (12). The adaptive constrained motion/force controller (16), with parameter update law (18), and joint torque sliding controller (20) stabilize the system and achieve global asymptotic convergence, i.e.*

$$\lim_{t\to\infty} e_p = 0, \quad \lim_{t\to\infty} e_f = 0 \quad \text{and} \quad \lim_{t\to\infty} e_t = 0$$

**Proof**: To prove the theorem, we first introduce the important lemma from [18],

**Lemma 1** *[18] Suppose that a symmetric matrix $Q$ is partitioned as:*

$$Q = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{12}^T & Q_{22} \end{bmatrix}$$

*where $Q_{11}$ and $Q_{22}$ are square. The matrix $Q$ is positive definite if and only if $Q_{11}$ is positive definite and $Q_{22} > Q_{12}^T Q_{11}^{-1} Q_{12}$.*

Choosing the Lyapunov function candidate as

$$\begin{aligned} V &= \frac{1}{2}S^T A(r)S + \frac{1}{2}\tilde{P}^T\Gamma\tilde{P} + \frac{1}{2}S_f^T k_{fD}S_f \\ &\quad + \frac{1}{2}s_t^T I_{ts}s_t + e_t^T\Lambda_t^T K_{tD}e_t > 0 \end{aligned} \quad (23)$$

the differentiation of $V$ is

$$\begin{aligned} \dot{V} &= S^T A\dot{S} + \frac{1}{2}S^T\dot{A}S + \tilde{P}^T\Gamma\dot{\tilde{P}} + S_f^T k_{fD}\dot{S}_f \\ &\quad + s_t^T I_{ts}\dot{s}_t + 2e_t^T\Lambda_t^T K_{tD}\dot{e}_t \end{aligned} \quad (24)$$

Using the fact that $(\dot{A}-2B)$ is skew symmetric and equation (17),

$$\begin{aligned} S^T A\dot{S} + \frac{1}{2}S^T\dot{A}S &= S^T(A\dot{S} + BS) \\ &= S^T(T^T e_t + Y_r\tilde{P} - T^T K_D TS - K_{fD}\dot{S}_f) \end{aligned} \quad (25)$$

together with equation (21), we have

$$\begin{aligned} \dot{V} &= S^T T^T e_t - S^T T^T K_D TS - S^T K_{fD}\dot{S}_f + S_f^T k_{fD}\dot{S}_f \\ &\quad - s_t^T B_{ts}s_t - \dot{e}_t^T K_{tD}\dot{e}_t - e_t^T\Lambda_t^T K_{tD}\Lambda_t e_t \\ &\quad + \tilde{P}^T(Y_r{}^T S + \Gamma\dot{\tilde{P}}) + s_t^T(w - k_t sgn(s_t)) \end{aligned} \quad (26)$$

With the parameter update law (18) and equality $S^T K_{fD}\dot{S}_f = S_f^T k_{fD}\dot{S}_f$,

$$\begin{aligned} \dot{V} &= -S^T T^T K_D TS + S^T T^T e_t - s_t^T B_{ts}s_t - \dot{e}_t^T K_{tD}\dot{e}_t \\ &\quad - e_t^T\Lambda_t^T K_{tD}\Lambda_t e_t + s_t^T(w - k_t sgn(s_t)) \\ &= -\begin{bmatrix} (TS)^T & e_t^T \end{bmatrix} Q \begin{bmatrix} TS \\ e_t \end{bmatrix} - s_t^T B_{ts}s_t - \dot{e}_t^T K_{tD}\dot{e}_t \\ &\quad + s_t^T(w - k_t sgn(s_t)) \end{aligned} \quad (27)$$

where

$$Q = \begin{bmatrix} K_D & -\frac{1}{2}I \\ -\frac{1}{2}I & \Lambda_t^T K_{tD}\Lambda_t \end{bmatrix}$$

By lemma 1, for matrix $Q$ to be a positive definite matrix, the requirement is

$$\Lambda_t^T K_{tD}\Lambda_t > \frac{1}{4}K_D^{-1}$$

Since $K_{tD}$, $\Lambda_t$ and $K_D$ are diagonal matrices of controller gains, we can choose them properly to meet the requirement, such that $Q > 0$. The last term in (27)

$$\begin{aligned} s_t^T(w - k_t sgn(s_t)) &\leq (\| s_t^T w \| - \| s_t^T k_t \|) \\ &\leq \| s_t \| (\varrho - \lambda_{k_t}^{min}) \end{aligned} \quad (28)$$

To meet the requirement of the sliding condition, the gain $k_t$ is selected such that the minimum eigenvalue $\lambda_{k_t}^{min}$ satisfies

$$\lambda_{k_t}^{min} > \varrho + \epsilon \quad (29)$$

Then

$$\dot{V} < -\begin{bmatrix} (TS)^T & e_t^T \end{bmatrix} Q \begin{bmatrix} TS \\ e_t \end{bmatrix} - s_t^T B_{ts}s_t - \dot{e}_t^T K_{tD}\dot{e}_t - \epsilon \| s_t \| \quad (30)$$

Thus, the entire system is stable in the sense of Lyapunov. We have $\tilde{P}$ bounded, $S \in L_\infty \cap L_2$ and $S_t \in L_\infty \cap L_2$. From equation (19), it immediately follows that

$$\lim_{t\to\infty} e_t = 0$$

In equation (17), we should notice that $\dot{v}_r = [\dot{v}_{r1}^T, -\dot{S}_f^T]^T$, $\dot{v}_{r1} = \ddot{r}_{pd} - k_p\dot{e}_p$. Rewrite the equation as

$$\begin{aligned} &A\begin{bmatrix} \dot{S}_p \\ 0 \end{bmatrix} + A\begin{bmatrix} 0 \\ \dot{S}_f \end{bmatrix} + (\hat{A} - A)\begin{bmatrix} 0 \\ \dot{S}_f \end{bmatrix} + K_{fD}\dot{S}_f = \\ &\quad -BS + (\hat{A} - A)\begin{bmatrix} \dot{v}_{r1} \\ 0 \end{bmatrix} + (\hat{B} - B)v_r + (\hat{W} - W) \\ &\quad + T^T e_t - (T^T K_D T)S \end{aligned} \quad (31)$$

partition A as $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, and denote

$$A_f = \begin{bmatrix} \hat{a}_{11} & \hat{a}_{12} \\ \hat{a}_{21} & \hat{a}_{22} + k_{fD} \end{bmatrix},$$

$$(\hat{A} - A)\begin{bmatrix} \dot{v}_{r1} \\ 0 \end{bmatrix} + (\hat{B} - B)v_r + (\hat{W} - W) = Y_{r1}(r, \dot{r}, v_r, S_f, \dot{v}_{r1})\tilde{P},$$

we have

$$A\begin{bmatrix} \dot{S}_p \\ 0 \end{bmatrix} + A_f\begin{bmatrix} 0 \\ \dot{S}_f \end{bmatrix} = -BS + Y_{r1}\tilde{P} + T^T e_t - (T^T K_D T)S \quad (32)$$

The RHS of equation is a bounded function (say, $f_{bounded}$) as $S$, $\tilde{P}$, $B$ and $Y_{r1}$ are bounded, $e_t \to 0$. Matrices $A$ and $A_f$ are nonsingular and bounded. Multiplying equation (32) from left by $[0, I]A^{-1}$, and denoting $A^{-1}A_f = \bar{A} = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix}$, we obtain

$$\alpha_{22}\dot{S}_f = [0, I]A^{-1} f_{bounded}$$

Since $\alpha_{22}$ is nonsingular as $A^{-1}A_f$ is nonsingular, $\dot{S}_f$ is bounded. Similarly, multiplying equation (32) from left



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by $[I,0]A_f^{-1}$, it can be shown $\dot{S}_p$ is bounded. Thus, $\dot{S}$ is bounded. It implies that $S$ is uniformly continuous, and $S \to 0$, which in turn implies $e_p \to 0$, $\dot{e}_p \to 0$ and $S_f \to 0$.

To show $e_f \to 0$, we need to show $\dot{S}_f \to 0$, or $\dot{S} \to 0$. Differentiating equation (32), gives

$$A\begin{bmatrix} \ddot{S}_p \\ 0 \end{bmatrix} + A_f \begin{bmatrix} 0 \\ \ddot{S}_f \end{bmatrix} = \dot{A}\begin{bmatrix} \dot{S}_p \\ 0 \end{bmatrix} + \dot{A}_f \begin{bmatrix} 0 \\ \dot{S}_f \end{bmatrix}$$
$$-B\dot{S} - \dot{B}S + Y_{r1}\dot{\tilde{P}} + \dot{Y}_{r1}\tilde{P} + T^T\dot{e}_t + \dot{T}^T e_t$$
$$-(T^T K_D T)\dot{S} - 2T^T K_D \dot{T} S \qquad (33)$$

Since $S, e_t$ and $\dot{\tilde{P}}$ converge to zero, $\tilde{P}$ and $\dot{S}$ are bounded, which in turn implies that $r, \dot{r}, \ddot{r}$ are bounded as desired trajectory are bounded, together with $r_{pd} \in C^3$, $\dot{Y}_{r1}, \dot{A}, \dot{A}_f, B$ and $T$ are bounded, hence, the RHS of equation is bounded. With the similar processing as above, it can be shown $\ddot{S}_f$ and $\ddot{S}_p$, i.e. $\ddot{S}$ are bounded, so that $\dot{S}$ is uniformly continuous and $\dot{S} \to 0$. The proof of the theorem is thus completed. □

In the proof, it has been shown that motion and force tracking errors can converge simultaneously to zero without persistent excitation conditions which are usually difficult to satisfy in practice. Distinct from the control method of [15], the proposed control scheme has a relatively simple formulation with PI type force feedback control function, and the restriction that the force feedback gain must be very small is not required. The nonsingularity of $A_f$ can be ensured by modifying the parameter update law with the reseting method [19, 20].

The control law (20) contains a switching function $k_t sgn(s_t)$, which may lead to high frequency control chattering. To remedy this problem, a continuous saturation function $sat(s_t)$ can be used to replace $sgn(s_t)$ [21].

## 4. SIMULATION EXAMPLES

Consider the two-link flexible joint robot manipulator in a vertical plane moving along a horizontal line, as shown in Figure 1.

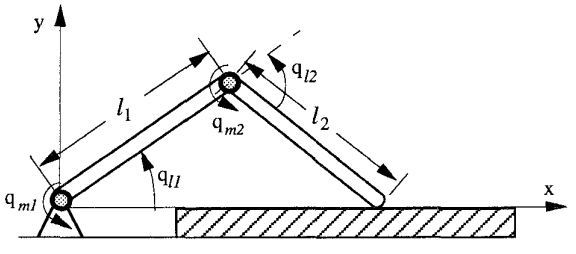



Figure 1: The schematic diagram of a robot manipulator and the constraint

For simplicity, assume $l_1 = l_2 = 0.25m$. The dynamic model of the rotor subsystems with joint torque sensors is given as follows:

$$\gamma_1(\gamma_1+1)J_{m1}\ddot{q}_{m1} + \gamma_1(\gamma_1+1)b_{m1}\dot{q}_{m1} + \tau_{s1} = u_1$$
$$\gamma_2(\gamma_2+1)J_{m2}\ddot{q}_{m2} + \gamma_2(\gamma_2+1)b_{m2}\dot{q}_{m2} + \tau_{s2} = u_2$$

and the corresponding link dynamics is:

$$H_{11}\ddot{q}_{l1} + H_{12}\ddot{q}_{l2} - h\dot{q}_{l2}^2 - 2h\dot{q}_{l1}\dot{q}_{l2} + G_1 = \tau_{s1} + f_1$$
$$H_{21}\ddot{q}_{l1} + H_{22}\ddot{q}_{l2} + h\dot{q}_{l1}^2 + G_2 = \tau_{s2} + f_2$$

The constraint is expressed in task space $X = [x, y]^T$ as

$$\Phi(X) = y = 0$$

With the proposed control scheme, if we choose $r_p = x$, and $r_f = \Phi(X) = y = 0$, then

$$J_q = J_X J = J \text{ as } J_X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

and we have $T = J_q^{-1} = J^{-1}$. The contact force can be determined to be

$$F_n = [0,1]^T\lambda, \qquad F_t = [1,0]^T\mu sgn(\dot{x})\lambda$$

and just as indicated by property 3,

$$T^T J^T F_n = [0,1]^T\lambda.$$

The constraint can also be expressed in the generalized joint space as

$$\Phi(X(q_l)) = q_{l2} + 2q_{l1} = 0$$

If we choose $r_p = q_{l1}$, and $r_f = \Phi(q_l) = q_{l2} + 2q_{l1} = 0$, then

$$J_q = \frac{\partial r(X(q_l))}{\partial q_l} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad T = J_q^{-1} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$$

and

$$T^T J^T F_n = \begin{bmatrix} 0 \\ l_2 c_{12} \end{bmatrix}\lambda$$

Because of the space limitation, only the simulation results with $r = [q_{l1}, 0]^T$ are shown in Figure 2. The desired trajectories are given as

$$r_{pd} = 40 + 20cos(0.5\pi t)\ (degree)$$
$$\lambda_d = 20 + 10cos\pi t\ (N)$$

It is illustrated that the system is stable and has good tracking performance.

## 5. CONCLUSION

In this paper, motion and force control of flexible joint robots in constrained motion is considered. A unified constrained dynamic model is derived with three important properties, which are essential in the controller design. A new adaptive and sliding control scheme is developed assuming that all system parameters including the joint flexibility values are unknown except for some of their bounds. The proposed control scheme, based on two-stage control strategy, consists of an adaptive constrained motion/force controller and a joint torque sliding mode controller. Distinct from the previous work in literature, the proposed controller with PI type force feedback control mechanism has a relatively simple formulation and has no restriction on force feedback gain. The tracking performance of the system is enhanced. It has been shown that with the proposed control scheme, motion, force and joint torque tracking errors converge simultaneously to zero without persistent excitation, which has not been, to our knowledge, proved by others for flexible joint robots. Simulation results illustrate the effectiveness of the proposed control method.



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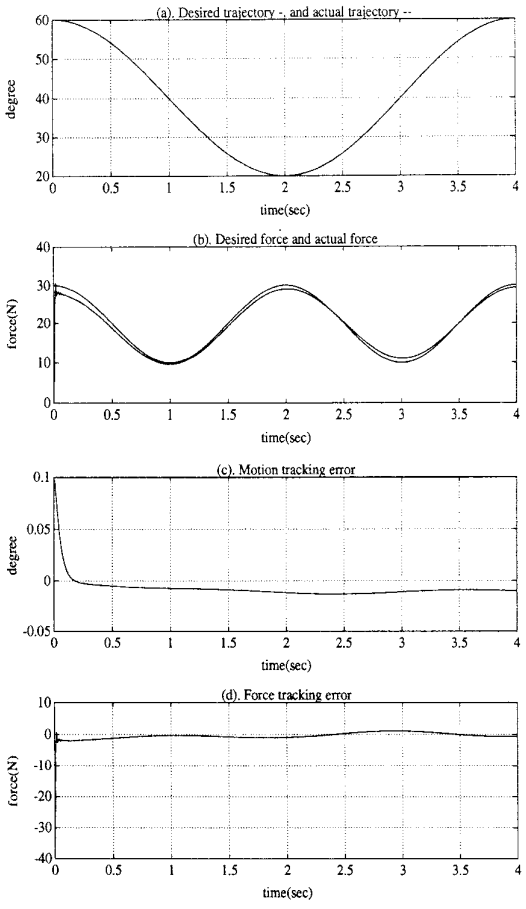



Figure 2: The response of the adaptive and sliding control system

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